

Investigating Line-Following Robot Navigation through Variance-Based Sensor Switching: A Hybrid Approach to Optimizing Sensor Performance

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Abstract—Navigational accuracy is crucial for the efficiency of line-following robots in industrial settings, yet the integration of highly accurate sensors often involves significant costs and complexity. We propose an innovative, cost-effective switching mechanism that dynamically selects between analog and digital sensor arrays based on real-time variance assessments, optimizing sensor performance without additional hardware. This approach utilizes the distinct advantages of both sensor types, the high-resolution data from analog sensors for precise control and straightforward binary outputs from digital sensors for reliable navigation. While our experiments demonstrate this hybrid sensor system enhances sensor reading accuracy and adapts well to varying track conditions, we found that the switching mechanism introduces instability, resulting in less smooth navigation and deviation from the intended line. This indicates a trade-off between sensor accuracy and operational smoothness, highlighting a critical area for further refinement to improve overall system performance.

I. INTRODUCTION

The role of line-following robots in manufacturing sectors has increased significantly, becoming pivotal in optimizing the logistics and transportation of goods within factory environments. These robots, equipped with advanced sensors and autonomous navigation capabilities, replace traditional, labor-intensive methods such as forklifts and cranes, which are not only high-maintenance but also pose significant safety risks to workers. Line-following robots offer a reliable and efficient solution by navigating predefined paths marked on factory floors, using a combination of proximity sensors for collision avoidance and vision sensors [1].

In practical applications, line-following robots can benefit from using both analog and digital sensors. Analog sensors provide high-resolution data and can detect slight variations in line position, which is crucial for fine-tuning the robot's trajectory. Digital sensors offer simplicity and robustness, making them suitable for clear, straightforward line detection. By incorporating both types of sensors, a line-following robot can effectively use the high-resolution data from the analog sensor for rigorous control and the binary signals from the digital sensor for straightforward decision-making processes. This dual-sensor approach enhances the robot's ability to follow lines accurately and adapt to varying conditions on the factory floor.

Line-following robots utilize infrared sensors to detect lines on the ground [2]. These sensors operate by emitting IR light and measuring the light that reflects back. The 3pi+

robot is equipped with five downward-facing line sensors located on the underside along the front edge, which are used to distinguish between light and dark surfaces using infrared (IR) technology. Each sensor comprises an IR LED and a phototransistor to detect reflected light.

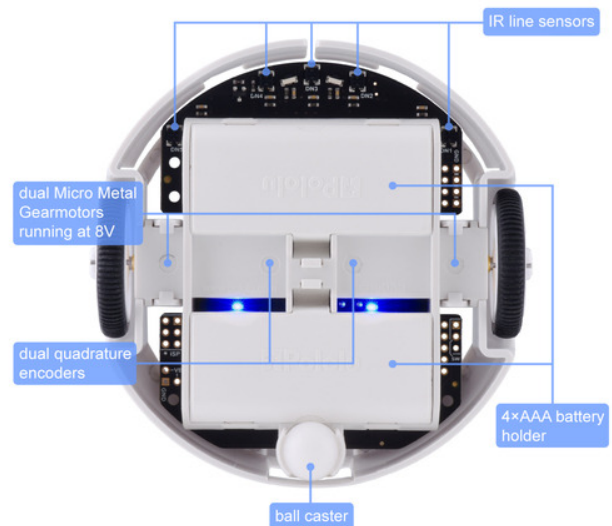


Fig. 1. Visual Representation of sensors on a 3pi+ Robot [3]

Analog Sensors

Analog IR sensors provide a continuous output signal that varies directly with the intensity of the reflected light. As a line-following robot approaches a black line, the output voltage of the analog sensor decreases proportionally to the amount of light absorbed by the line. This continuous signal allows for subtle adjustments in the robot's path, ensuring smoother line-following. However, analog signals are susceptible to noise and may require additional filtering or signal conditioning to maintain accuracy.

Digital Sensors

Digital IR sensors, on the other hand, convert the analog input from the IR receiver into a binary output. This is achieved by setting a threshold level on the sensor; when the reflected light intensity falls below this threshold (indicating a black line), the output is set high (1), and when it is above, the output is low (0). This binary output simplifies the processing requirements for the robot's microcontroller or digital system, making the setup more robust against noise and easier to interface with digital components.

Calibration is vital for these sensors to ensure they perform accurately across different environments and lighting conditions, involving setting thresholds for light and dark detection. Proper calibration and adjustments are essential for the robot to perform optimally in diverse tasks and settings.

A. Hypothesis Statement

Our proposal focuses on developing a variance-based switching mechanism for line-following robots that dynamically selects between analog and digital sensor arrays. This selection is based on which sensor array shows the lower variance at any given time. By leveraging the distinct advantages of both sensor types and minimizing their potential weaknesses, this method aims to enhance the robot's operational efficiency and adaptability.

We hypothesize that autonomous switching between analog and digital sensor readings based on variance will lead to improved performance in line-following robots compared to using either sensor type alone. This improvement is expected in terms of both accuracy and adaptability in varying environmental conditions. To validate our hypothesis, we plan to conduct a series of controlled experiments.

If successful, this approach could significantly enhance the robustness and flexibility of line-following robots, making them more adept at operating in diverse and challenging industrial environments. This would not only prove the efficacy of variance-based sensor switching but could also set a precedent for the broader application of hybrid sensor systems in automated technology and robotics. By integrating this innovative sensor technology, we anticipate improvements in manufacturing workflows, increased automation capabilities, and higher precision in tasks that require diligent movement control.

II. IMPLEMENTATION

A. Sensor Calibration Procedures

The calibration process is the first stage in our process towards achieving autonomous line-following. We carefully execute this step to condition the sensors to the robot's operating environment. Calibration aligns the analog and digital sensor outputs to a standardized range, thereby enhancing the robot's sensitivity to the contrast between the black line and its white background. Initially, we expose the sensors to a controlled sequence of the two contrasting surfaces. This setup is essential in capturing the extremities of sensor values, which are then utilized to normalize the readings. We proceed with the calibration in a methodical manner, iterating over the track to ensure that each sensor's range encompasses the full spectrum of expected values.

1) Calibration Process:

a) *Initialisation:* Upon startup, we engage the robot in a pre-programmed routine where it rotates over the contrasting surfaces of the track. As the robot spins, each sensor captures a series of readings. For the analog sensors, we collect voltage variations, while the digital sensors measure time intervals indicative of reflectivity. Voltage variations refer to the fluctuations in output voltage of analog sensors as they detect different surface colors and textures. These variations are critical as they directly correlate with the sensor's ability to discern the line from the surrounding

floor. We record the maximum and minimum values from the collected data, which represent the sensor's response to the black line and the white background, respectively. Using these extremities, we compute the normalization parameters—scale and offset.

b) *Normalisation:* These parameters are unique to each sensor and are determined as follows:

- The offset is set to the minimum value, essentially zeroing the sensor readings at the lowest point.
- The scale factor is the reciprocal of the range (maximum minus minimum), transforming the sensor readings to a [0,1] scale.

c) *Application:* The calculated offset and scale are then applied to all subsequent sensor readings, effectively normalizing them and making the sensors agnostic to environmental variances such as lighting changes or surface inconsistencies.

Algorithm 1 Sensor Reading Calibration

Input:

Output: Normalized Readings

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1: for  $i$  in [analog, digital] do
2:   Offset[ $i$ ] = MinReading[ $i$ ]

   Scale[ $i$ ] =  $\frac{1}{\text{MaxReading}[i] - \text{MinReading}[i]}$ 

3:   NormalizedReading[ $i$ ] = (RawReading[ $i$ ] - Offset)
   * Scale
4: end for

```

B. Variance Calculations

We carefully assessed the sensor reliability, which is where the concept of variance becomes indispensable. Variance allowed us to gauge the consistency of our sensors' readings, a fundamental property for ensuring that the robot's perception of its path remains steadfast. A smaller variance implies that the sensor readings are tightly clustered around the mean, signifying a high degree of reliability and stability in the measurements obtained from trailing the black line. Variance in this context is defined as the statistical variance among readings from all sensors at a given time

Algorithm 2 Variance Results

Input: **Output:** Variances

```

1: for sensors  $i = 1$  to 5 do
2:   Collect  $N$  sensor readings
3:   Compute Mean of  $N$  sensor readings
4:   Mean[ $i$ ] =  $\frac{\sum_{j=1}^N R[i][j]}{N}$ 
5:   Compute Variance for sensor  $i$ 
6:   Variance[ $i$ ] =  $\frac{\sum_{j=1}^N (R[i][j] - \text{Mean}[i])^2}{N}$ 
7: end for

```

In implementing these steps, we not only ensure that our robot's line detection is robust against transient anomalies, but we also establish a framework for predictive maintenance. By monitoring the variance over time, we can preemptively identify and rectify sensor drifts or failures, thus maintaining the integrity of the robot's orientation.

C. Sensor Measurement Selection Logic

Our robot's ability to discern the most accurate path relies heavily on selecting the best set of sensors for line-following. This selection is critical because each type of sensor, either analog or digital, responds differently to specific environmental conditions. To ensure the most stable data guides the robot, we implemented a logic that evaluates the consistency of sensor readings through variance.

1) *Selection Process*: The robot continuously computes the variance for the readings from both sensor arrays. Leveraging the calculated variances, our logic determines which sensor array provides the most consistent data. The robot then selects the sensor array with the lower variance, reflecting more reliable and stable readings.

Algorithm 3 Sensor Method Selection

Input:	Output: Method
1: if <i>analogVariance</i> < <i>digitalVariance</i> then	
2: Use <i>Analog</i> Method	
3: else	
4: Use <i>Digital</i> Method	
5: end if	

D. Line-following Control Logic

We have designed a system that employs weighted sensor measurements to determine the magnitude and direction of corrective actions necessary to stay true to the course. Weights are assigned to sensor inputs based on their relative positions on the robot. Sensors closer to the line contribute more significantly to the input calculation. The corrective actions are derived from a proportional control system where the response is scaled relative to the deviation. The greater the deviation, the more significant the adjustment made to the motor speeds.

Each sensor array provides a set of readings that reflect the robot's position relative to the line. These readings are not merely aggregated but weighted according to their position on the sensor array, with an emphasis on those readings that indicate the robot's deviation from the line. The weighted measurements form the basis for calculating the deviation. This value is then used to determine the corrective actions required for the motors, allowing the robot to adjust its trajectory. The control algorithm's effectiveness lies in its ability to interpret sensor data as actionable intelligence. It takes into account the nature and degree of deviation and translates it into precise motor power (PWM) signals.

1) Mathematical Expression of Motor Control:

- $W = \frac{\text{WeightedRightSensor} - \text{WeightedLeftSensor}}{\text{WeightedRightSensor} + \text{WeightedLeftSensor}}$
- $\text{LeftPWM} = \text{BiasPWM} + (\text{MaxTurnPWM} \cdot W)$
- $\text{RightPWM} = \text{BiasPWM} - (\text{MaxTurnPWM} \cdot W)$

III. EXPERIMENTAL METHODOLOGY

The purpose of this experimental study is to rigorously evaluate the performance of three distinct line-following methods for autonomous robots: analog, digital, and a variance-based switching mechanism. The goal is to determine the optimal line-following strategy under varying

track conditions, thereby contributing to the advancement of navigation algorithms for autonomous systems.

We hypothesize that the variance-based switching method will outperform the individual analog and digital methods by dynamically selecting the sensor array (analog or digital) that demonstrates the most reliable readings in real-time. This approach is expected to improve the robot's line-following accuracy and reduce deviation from the centre line, and enhance its adaptability to environmental fluctuations. From experimentation, we gather data and key metrics such as smoothness and deviation from the line as our robot line follows. From these, we can evaluate the efficacy of each method and compare it with our switching line-following to gain insight into the applicability of the proposed improvement.

A. Experimental Design

1) Track Configurations: Isolated track parts

- **Straight Line**: A straight segment isolates the core line-following behaviour and minimizes the influence of turning manoeuvres.
- **Curve**: A curved segment introduces a specific challenge, testing how well each method handles the continuous directional change.

Testing on isolated track segments enables focused analysis of how the line-following methods handle specific challenges, aiding in identifying where and why performance differs. Comparing performance on identical track segments between methods provides clear insight into relative strengths and weaknesses.

2) Trials: 10 Testing Samples

Each line-following method (analog, digital, switching) will undergo 10 trials on both the straight and curved track segments. This number of trials is significant to facilitate robust statistical analysis, ensuring that observed differences in performance are genuine and not due to random variations.

3) Variables: **Controlled Variables**

- **Environment**: Consistent lighting and track surface.
- **Battery Power**: Ensure the robot is fully charged before each trial.
- **Sensor Calibration**: Calibration procedure applied before each run.

Independent Variable

- The line-following method (analog, digital, switching)

Dependent Variables

- **Theta Values** - Theta values of the robot using kinematics as the robot performs line-following. These values will be used to calculate key metrics such as deviation from the line and smoothness of the line-following.
- **Sensor Variances** - Variance calculated for each sensor array during line-following. This metric can be used to determine the stability and reliability of each sensor reading method.

B. Experimental Procedure

1) Setup and Calibration

- a) *Environment Preparation*: Ensure consistent lighting conditions and use the same track segments for all trials.

- b) *Robot Power*: Fully charge the robot's battery.
- c) *Sensor Calibration*:
 - Place the robot on the designated calibration area, with its central sensor positioned over the line.
 - Execute the calibration routine, spinning the robot to capture and normalize the full range of expected sensor readings.

2) Trial Execution

- a) *Method Selection*: Choose the line-following method (analog, digital, or switching).
- b) *Robot Placement*: Align the robot in the middle and start of the line and power on.
- c) *Data Collection*: Once the robot has been powered on, calibration occurs and the robot spins in place. From here, realign the middle sensor with the starting point of the track part and simultaneously record theta values and sensor variances.
- d) *Completion*: Once the robot has detected the end of the line, it will stop.
- e) *Data Storage*: Store the recorded values in a CSV file for analysis.

C. Discussion of Key Metrics

1) Deviation from Centreline

Calculating deviation directly measures our primary goal of effective line-following. Deviation offers us a clear way to compare performance between the different methods. This metric is valuable to us given our approach of testing on parts of a track with consistent width lines. However, comparisons can be skewed if we tested on track parts with different widths. As well as this, we are unable to capture rapid oscillations around the line as the robot follows using solely deviation as a metric.

2) Smoothness from Theta values

These theta values are readily available to be obtained from our implementation of kinematics and allow us to reveal how smoothly the robot follows, excessive oscillations would give rise to large fluctuations in theta. Spikes in our theta plot might indicate overcorrection or problems with the sensors in specific track sections. Theta value alone primarily reveals turning behaviour and not absolute deviation from the line. Given that we are testing on curves, theta changes more requiring a more careful comparison between track sections.

3) Variances

By calculating and measuring the variance levels of the sensors, we are able to gain direct insight into the fluctuation and noise levels of the readings. Consistently high variances for certain methods can reveal a lack of stability in its sensor readings whilst low variances lead us to believe that the line-following method is working with the highest accuracy in terms of understanding the environment. Variances alone do not explicitly inform us of the performance of line-following for its method as it can't tell us how far the robot is from the line. To assess the consistency and reliability of sensor readings across different methods, the Interquartile Range (IQR) of sensor variances will

be used as a key metric. The IQR represents the middle 50% of the data points, providing a measure of statistical dispersion and a clearer view of data clustering without the influence of outliers.

The collation of these 3 key metrics gives us an in-depth understanding of the capabilities of the robot and allows us to determine our hypothesis that the use of switching between sensor reading methods, based on the lowest variance, can lead to improved line-following performance compared to solely using analog or digital sensor reading methods. The combination of metrics can also highlight areas for algorithmic improvement. For example, a scenario where the robot achieves low deviation but demonstrates high smoothness variation and high sensor variance might indicate the robot manages to stay on course but with constant overcorrections and sensor noise might be causing the algorithm to make unnecessary adjustments. The analysis of performance solely using deviation from the line might suggest success but in tandem with our other metrics give a better view of the true ability of the robot.

IV. RESULTS

In Figures 1 and 2, we see a comparison of sensor variance for each line-following method on both track parts.

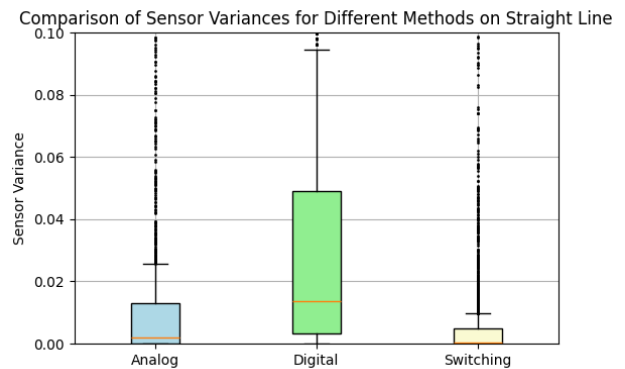


Fig. 2. Box Plot of Sensor Variances for Straight Line Tracking: Comparing Analog, Digital, and Switching Methods.

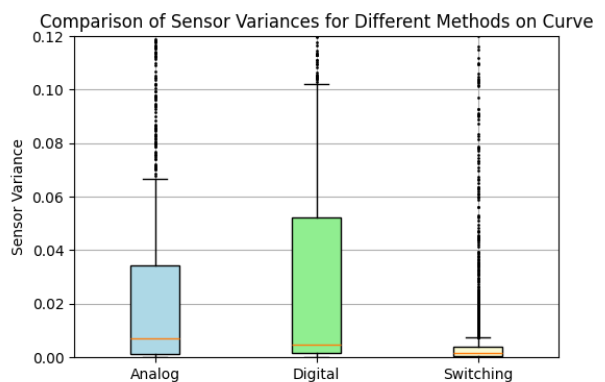


Fig. 3. Box Plot of Sensor Variances for each method on Curve

The digital method shows a large IQR whilst analog shows a much lower and switching shows a significantly

lower range. From this, we can determine that sensor variances are most consistent for the Switching method. Looking at the upper whiskers, as well as the median values (orange line) for each method, again we see that variances are higher for digital and variances are least for the switching method. Low variance indicates stability and precision in sensor readings and validates our calibration method. This provides a basis for our hypothesis that the switching method could lead to improved line performance given that this method seems to produce the lowest deviation in readings thereby giving a more consistent and reliable input to our weighted measurements for line-following.

However, this reliability is compromised by the presence of a significant amount of outliers (black dots) for the switching method. These outliers represent errant readings which counter the consistency demonstrated by the dimensions of the boxplot. The high number of outliers could be caused by various reasons such as instability in the switching mechanism itself. If the algorithm switches frequently between the methods, potentially because of a lack of strictness in the switching criteria, there could be overfitting to noise. Another reason could be potential discrepancies in the calibration values for the different methods. This could induce disparate readings when switching occurs. Overall this could imply that whilst the switching method has potential for high performance under normal conditions, the outliers mitigate the robustness of the system and suggest the switching algorithm may need refinement.

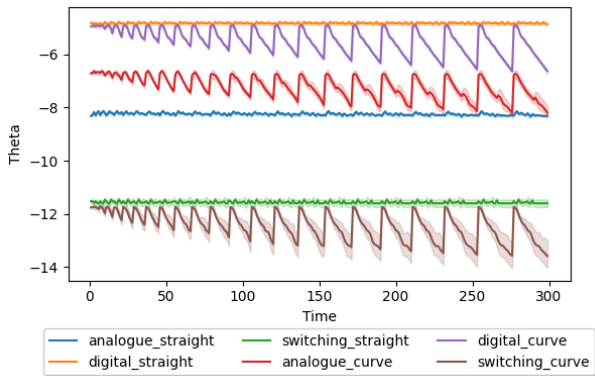


Fig. 4. Comparison of Theta Values Across Line-Following Methods: Illustrating the Theta values for Analog, Digital, and Switching methods over the first 300 seconds on both straight and curved track scenarios

Figure 3 shows the recorded Theta values for the first 300 seconds for all methods on straight and curve scenarios. As seen by the straight-line scenario, whilst all methods show minimal fluctuations in the theta values, the analog method seems to be the smoothest, indicative of a more controlled and steady line-following approach. The more jagged nature of the switching straight line scenario suggests a relatively smooth process but undergoing more corrections or deviations.

For the curve scenario, the analog method demonstrates the smoothest and most consistent fluctuations. The pattern of gradual increases and decreases in Theta again correspond with the methodical approach to line-following with predictable adjustments which can be interpreted as the method with the highest levels of smoothness. The switching curve shows the most significant changes in

fluctuations with the most erratic pattern. The frequent and abrupt changes in Theta indicate the line-following process is the least smooth and there is greater deviation from the line when following the curve. In an ideal line-following method, we would hope to see consistent and small fluctuations in Theta that mirror the geometry of the curve.

The less-than-ideal depiction of the switching method's Theta could be due to different factors impacting the smoothness again relating to the switching dynamics of the algorithm itself causing more frequent shifts in the robot's perceived optimal path. Overcorrections could be another factor contributing to the hindrance of line-following performance, particularly for the curve scenario. When performing line-following, there could be a delay in the algorithm's decision-making process that results in the robot overcorrecting and giving rise to erratic changes of Theta as the robot tries to maintain consistency.

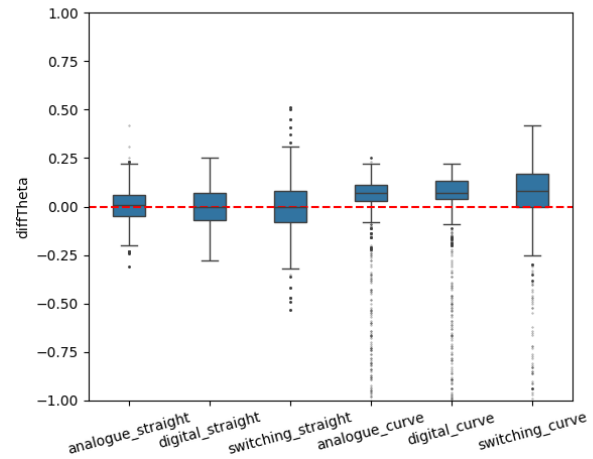


Fig. 5. Differential Theta Values for Line-Following Methods: Displaying the differences in Theta values from one measurement to the next for each line-following method during tests on straight and curved tracks

Figure 4 shows the distribution of differences in theta values from one timepoint to the next. For the straight line track part, the median shows consistently around zero for all methods. This is expected as we do not expect to see much bias in the turning rate from one reading to the next. The IQR is smallest for analog in both the straight and curve scenarios which implies a smoother line-following. Whilst the analog and digital method both show smaller IQR for the curve compared to straight line, the switching method's IQR does not decrease on curve. This could suggest that the adaptive mechanism does not improve line-following performance in terms of smoothness, and therefore reducing robustness hypothesised by using the switching method, this is also highlighted by the longer whiskers for the switching methods which tell us that there are a larger range of diffTheta values, thereby indicating larger deviations. A smaller length in whiskers as shown for the analog and digital curve can be interpreted as a smoother line-following with less unpredictable movement.

All methods have medians above the zero line for the curve scenario, this could signify a systematic bias in control response when following a curve and can suggest

all methods tend to over-steer or under-steer consistently. This can further be highlighted by the significant increase in outliers below the lowest whisker for all curve scenarios. These outliers can be justified by the fact that in experimentation, for all curve scenarios, the robot moved in an anti-clockwise direction. The negative diffThetas represent the robot turning left more as is in line with the anti-clockwise direction. The outliers can tell us that there was indeed over-steering to the left as the robot followed the curve.

V. DISCUSSION AND CONCLUSION

Our hypothesis posited that autonomous switching between analog and digital sensor readings based on variance would lead to improved performance in line-following robots compared to using either sensor type alone. This improvement was anticipated in terms of both accuracy and adaptability under varying environmental conditions. The results presented in the earlier sections offer a nuanced view of this hypothesis.

The experimental data (Figures 1 and 2) indicated that the switching mechanism, designed to choose the sensor array with the lower variance, indeed produced the most consistent sensor variances as observed in the box plots of both straight and curved track segments. This suggests that the switching strategy can enhance the stability of sensor readings, which is critical for precise line-following. However, while the lower variance supports the hypothesis, the high number of outliers in the switching method suggests issues with the switching mechanism's stability and highlights the need for refining the switching algorithm to handle these anomalies effectively.

The switching method did not unequivocally translate these into the most effective line-following performance. Particularly, the Theta values and their fluctuations (Figure 3) helped us understand the operational dynamics under different conditions and highlighted significant challenges. On curves, where stable and consistent navigation is crucial, The switching method displayed more erratic and jagged fluctuations compared to the analog method. This discrepancy between low sensor variances and actual navigation performance suggests that while the sensors provide reliable data, the method of integrating these into the control logic may not be optimal, leading to overcorrections or insufficient responsiveness in real-time adjustments.

The hypothesis was instrumental in revealing the complexities of sensor data integration into robotic control systems, highlighting the distinction between data reliability and control efficacy.

The implications of our study extend to the broader field of autonomous robotic navigation, particularly in environments requiring high precision and adaptability. While our results validate the potential of a variance-based switching mechanism to enhance sensor reliability, they also caution against an over-reliance on sensor data without adequate attention to the control algorithms that utilize this data.

For future studies and applications, it is recommended to explore more sophisticated integration techniques that not only consider the variance but also how quickly the variance changes, or perhaps a predictive model that anticipates future sensor readings based on past trends.

Refinement in the switching criteria, to reduce the frequency of switching, would be beneficial to improve the

stability of real-time response to the track. For instance, a threshold could be employed where switching only occurs if variance exceeds this threshold for a consistent period, rather than instantaneously.

Filtering techniques, like a Kalman filter, for the sensor data could be implemented to smooth out the inputs before they are used in control decisions. This could mitigate the frequency and impact of outliers thereby reducing the occurrence of erratic behaviour due to abrupt data changes. This is particularly important in environments with variable light or surface conditions and enhance the robustness of the system.

Using machine learning to refine the model and learn from the robot's previous data could allow us to dynamically adjust the control parameters based on current and past performance metrics. This could help again to manage control responses for both straight and curved paths.

The study revealed limitations in the experimental setup, such as the challenge of replicating consistent environmental conditions that could affect sensor readings. Future work should aim to include more dynamic environments in testing to better simulate real-world conditions, which would provide a more rigorous assessment of the switching mechanism's robustness.

In conclusion, the study demonstrated foundational success in enhancing sensor reliability through a switching mechanism, yet its real-world applicability in robotic navigation requires further development and refinement. The work conducted here paves a way for more integrated and adaptive control systems in robotics.

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